BASED ON LATEST OCTOBER PATTERN

Infosys Pseudo Code Questions

Topic: Strings & Character Manipulation

Question 01

```
Set Character array c = {'P', 'Q', 'R', 'S'}
Set Character result = c[2] + 32
display result
```

A) r

B) R

C) q

D) s

```
Set Character array c = {'P', 'Q', 'R', 'S'}
Set Character result = c[2] + 32
display result
```



Answer: A) r

Execution Dry Run

Array Init

`c` initialized with **'P', 'Q', 'R', 'S'** at indices 0 to 3.

Fetch Char

`c[2]` retrieves the character **'R'**.

ASCII Math

ASCII of 'R' is **82**. Calculation: **82 + 32 = 114**.

Result

ASCII 114 corresponds to lowercase **'r'**.

Question 02

```
Set String str = "Info"
Set String res = ""
Set Integer i = 0
while(i < 4)
do
res = res + str[i] + str[i]
i = i + 1
end-while
display res
```

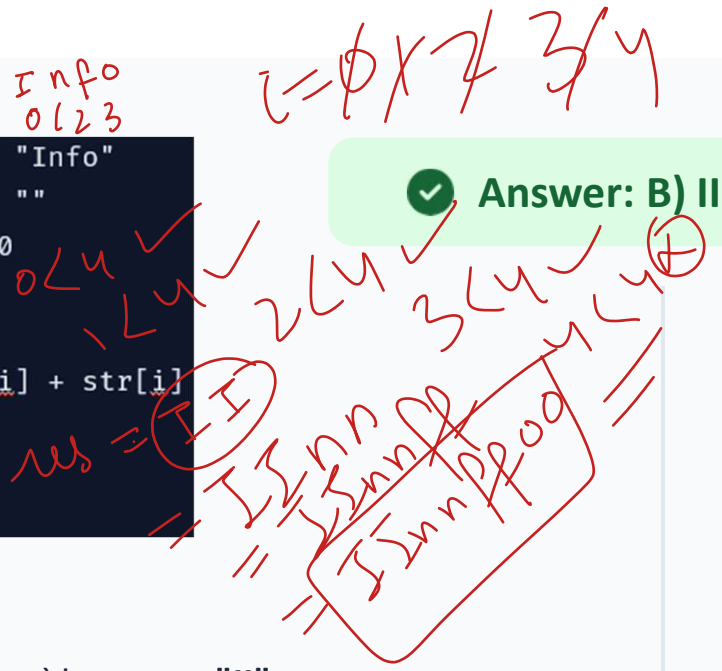
A) InfoInfo

B) IInnffoo

C) IIInnnooo

D) Error

```
Set String str = "Info"
Set String res = ""
Set Integer i = 0
while(i < 4)
do
res = res + str[i] + str[i]
i = i + 1
end-while
display res
```



Answer: B) IInnffoo

i=0: `str[0]` is 'I'. `res` becomes **"II"**.

i=1: `str[1]` is 'n'. `res` becomes **"IInn"**.

i=2: `str[2]` is 'f'. `res` becomes **"IInnff"**.

i=3: `str[3]` is 'o'. `res` becomes **"IInnffoo"**.

Loop Ends: Condition `4 < 4` is False.

Question 03

```
Set String s = "apple"
Set Integer n = 5
for i = 0 to n-1
step 1 do
if(s[i] == 'a' OR s[i] == 'e')
then do
s[i] = '*'
end-if
end-for
display s
```

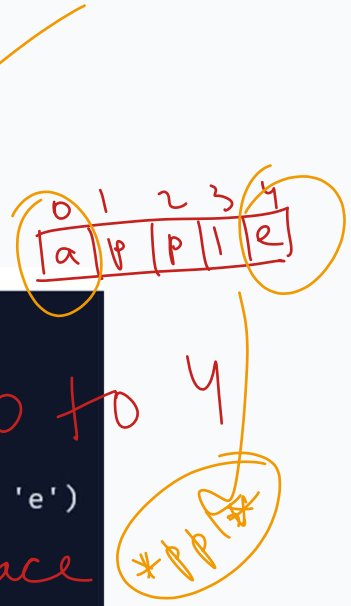
A) ale

B) *pple

C) apple*

D) *ppl*

Answer: D) *ppl*




```
Set String s = "apple"
Set Integer n = 5
for i = 0 to n-1
step 1 do
if(s[i] == 'a' OR s[i] == 'e')
then do
s[i] = '*'
end-if
end-for
display s
```

Iteration i=0

`s[0]` is 'a'. Condition is **True**. `s[0]` becomes '*'. String is ***pple**.

Iterations i=1, 2, 3

Characters are 'p', 'p', 'l'. Condition is **False** for all. No change.

Iteration i=4

`s[4]` is 'e'. Condition is **True**. `s[4]` becomes '*'. String is ***ppl***.

Question 04

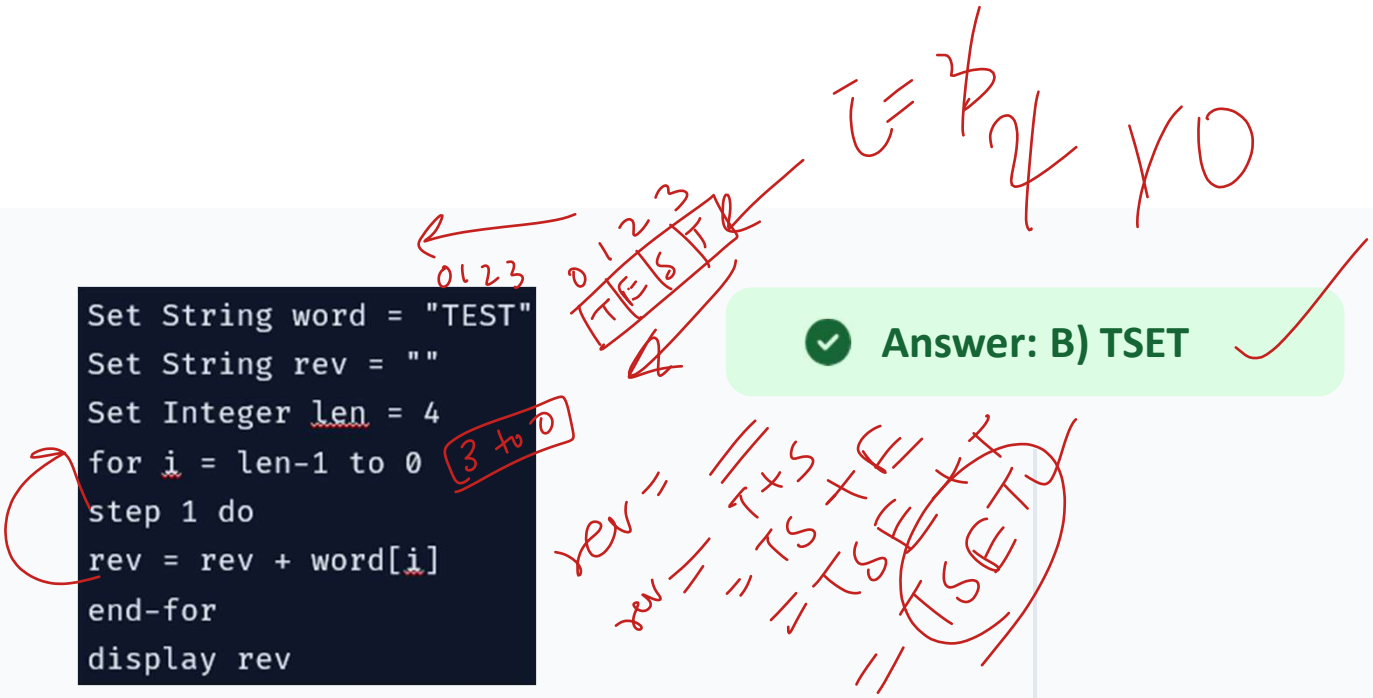
```
Set String word = "TEST"
Set String rev = ""
Set Integer len = 4
for i = len-1 to 0
step 1 do
rev = rev + word[i]
end-for
display rev
```

A) TEST

B) TSET

C) ETST

D) TTSE



```
Set String word = "TEST"
Set String rev = ""
Set Integer len = 4
for i = len-1 to 0
step 1 do
rev = rev + word[i]
end-for
display rev
```

Answer: B) TSET

i=3: `word[3]` is 'T'. `rev` = "" + 'T' = **"T"**.

i=2: `word[2]` is 'S'. `rev` = "T" + 'S' = **"TS"**.

i=1: `word[1]` is 'E'. `rev` = "TS" + 'E' = **"TSE"**.

i=0: `word[0]` is 'T'. `rev` = "TSE" + 'T' = **"TSET"**.

Question 05

```
Set String str = "banana"
Set Integer count = 0
for i = 0 to 5
step 1 do
if(str[i] == 'a')
then do
count = count + 1
end-if
end-for
display count
```

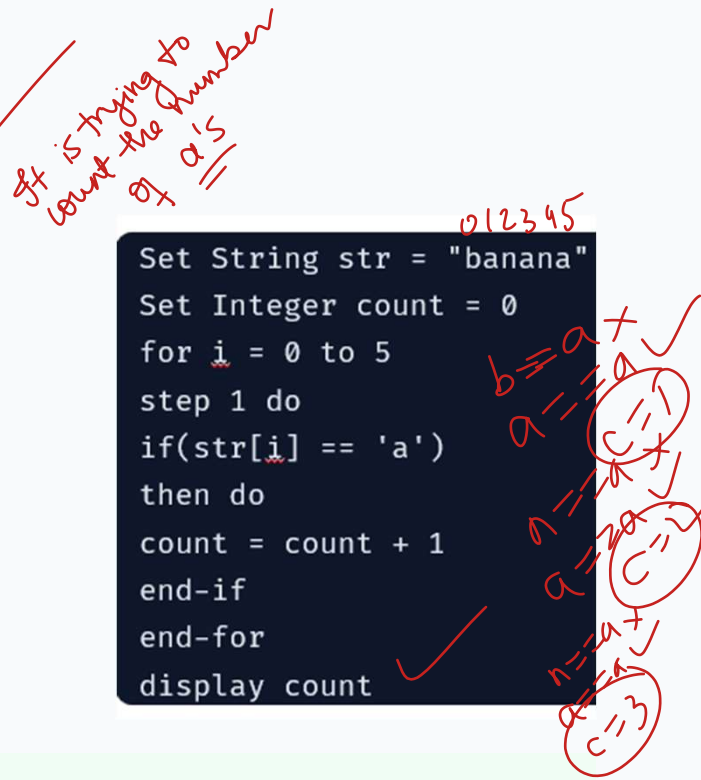
A) 2

B) 3

C) 4

D) 5

Answer: B) 3



Loop Iteration

Iterates through each character of **"banana"** from indices 0 to 5.

Condition Match

Target **'a'** is found at indices **1, 3, and 5**.

Final Count

The `count` variable increments **3 times**. Output is 3.

Question 06

```
Set String s = "PROGRAM"
Set String out = ""
Set Integer i = 2
while(i < 5)
do
out = out + s[i]
i = i + 1
end-while
display out
```

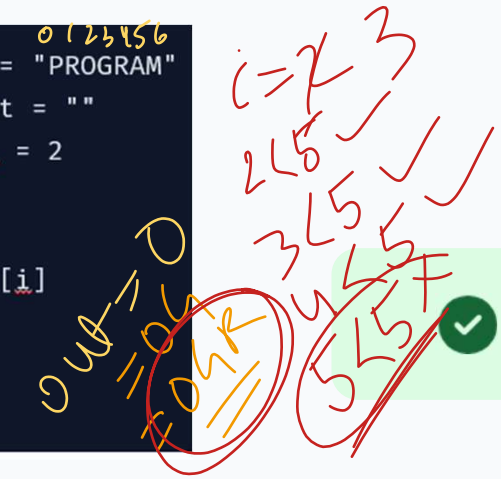
A) PRO

B) GRAM

C) OGR

D) ROG

```
Set String s = "PROGRAM"
Set String out = ""
Set Integer i = 2
while(i < 5)
do
out = out + s[i]
i = i + 1
end-while
display out
```



Answer: C) OGR

i = 2: `s[2]` is 'O'. `out` becomes **"O"**.

i = 3: `s[3]` is 'G'. `out` becomes **"OG"**.

i = 4: `s[4]` is 'R'. `out` becomes **"OGR"**.

Question 07

```
Set String str = "MADAM"
Set Integer left=0, right=4
Set Integer flag = 1
while(left < right)
do
if(str[left] != str[right])
then
flag = 0
break
left = left + 1, right = right - 1
end-while
display flag
```

A) 0

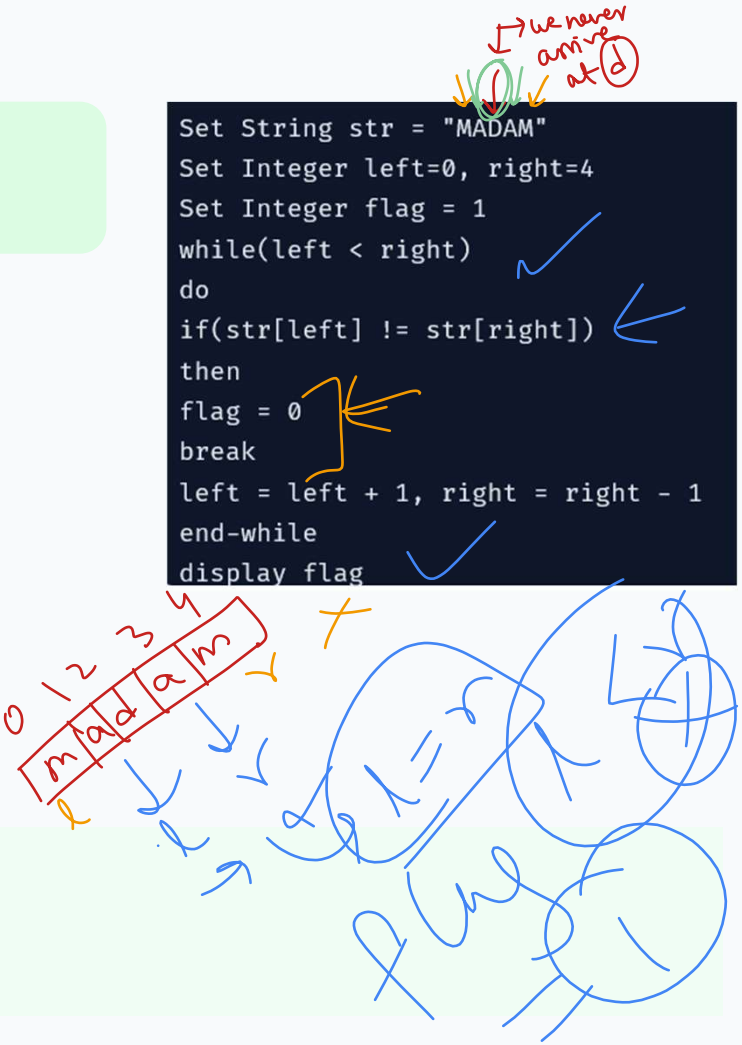
B) 1

C) MADAM

D) Error

Answer: B) 1

```
Set String str = "MADAM"
Set Integer left=0, right=4
Set Integer flag = 1
while(left < right)
do
if(str[left] != str[right])
then
flag = 0
break
left = left + 1, right = right - 1
end-while
display flag
```



Iteration 1

`str[0]` ('M') == `str[4]` ('M'). Match. `left` becomes 1, `right` becomes 3.

Iteration 2

`str[1]` ('A') == `str[3]` ('A'). Match. `left` becomes 2, `right` becomes 2.

Result

Loop ends. No mismatches found. `flag` remains **1**.

Question 08

```
Set Character ch = 'G'
Set Integer ascii_val = ch
if(ascii_val >= 65 AND ascii_val <= 90)
then
ascii_val = ascii_val + 32
end-if
display Character(ascii_val)
```

A) G

B) g

C) h

D) Error

Answer: B) g

ASCII Conversion

`ch` is 'G'. ASCII value of uppercase 'G' is **71**.

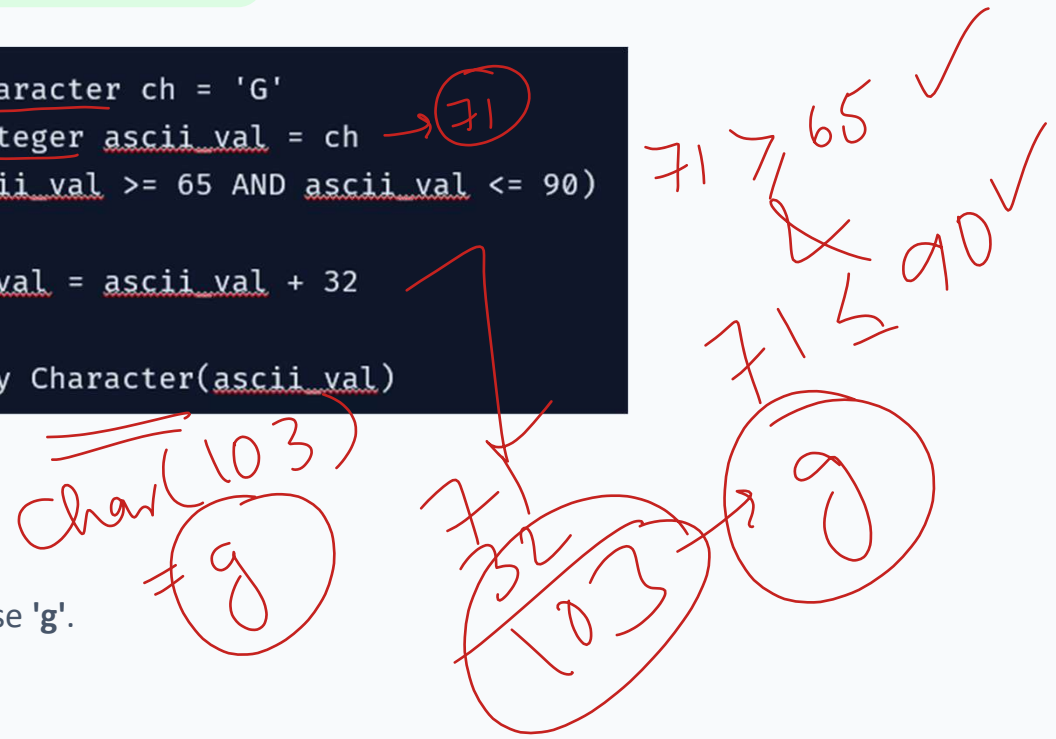
Uppercase Check

Condition `71 >= 65 AND 71 <= 90` is **True**.

Shift Value

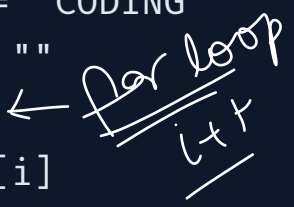
`ascii_val` updated: 71 + 32 = **103**. Casting 103 yields lowercase **'g'**.

```
Set Character ch = 'G'
Set Integer ascii_val = ch
if(ascii_val >= 65 AND ascii_val <= 90)
then
ascii_val = ascii_val + 32
end-if
display Character(ascii_val)
```



Question 09

```
Set String text = "CODING"
Set String res = ""
for i = 0 to 5
res = res + text[i]
i++
end-for
display res
```



A) CDI

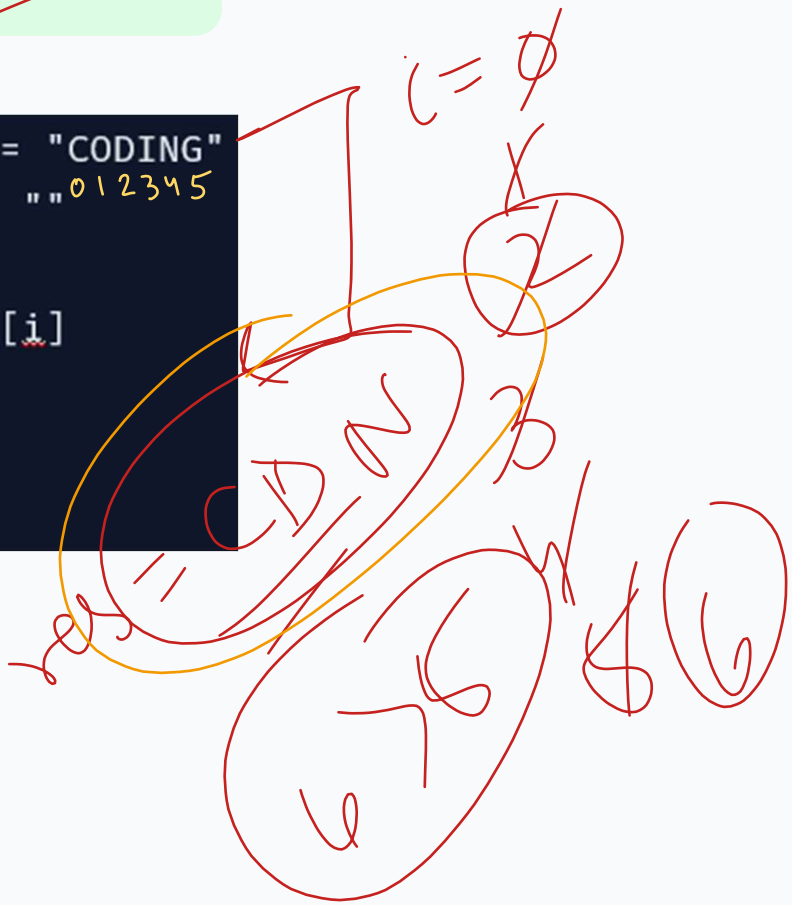
B) ODG

C) OIG

D) CDN

Answer: D) CDN

```
Set String text = "CODING"
Set String res = ""
for i = 0 to 5
res = res + text[i]
i++
end-for
display res
```



Iteration i=0

`text[0]` is 'C'. `res` becomes **"C"**.

Iteration i=2

`text[2]` is 'D'. `res` becomes **"CD"**.

Iteration i=4

`text[4]` is 'N'. `res` becomes **"CDN"**.

Question 10

```
Set String str1 = "INFOSYS"
Set String str2 = "TECH"
Set Integer val = Length(str1) - Length(str2)
display val * 2
```

A) 3

B) 6

C) 11

D) 22

Answer: B) 6

```
Set String str1 = "INFOSYS"
Set String str2 = "TECH"
Set Integer val = Length(str1) - Length(str2)
display val * 2
```

String Lengths

- Length of `str1` ("INFOSYS") = **7**
- Length of `str2` ("TECH") = **4**

Calculation

- `val` = 7 - 4 = **3**
- Display: `val * 2` -> 3 * 2 = **6**



End of Strings Practice Set

Keep practicing for your technical rounds!